AI-Based Algorithms Used in Smart AI-Based Food Nutrition Analysis

1. Traditional Machine Learning Algorithms

- Support Vector Machine (SVM)
- Random Forest
- K-Nearest Neighbors (KNN)
- Decision Trees
- Naïve Bayes

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2. Convolutional Neural Network (CNN) Architectures

- AlexNet
- VGG16
- VGG19
- GoogLeNet (Inception)
- ResNet
- DenseNet
- EfficientNet
- MobileNet
- NASNet

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3. Object Detection Algorithms

- R-CNN
- Fast R-CNN
- Faster R-CNN
- SSD (Single Shot MultiBox Detector)
- YOLOv3
- YOLOv4
- YOLOv5
- YOLOv7
- YOLOv8
- YOLOv9
- YOLOv10

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4. Food Segmentation Algorithms

- Fully Convolutional Network (FCN)

- U-Net
- Mask R-CNN
- DeepLabV3
- PSPNet
- SegFormer

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5. Portion Size Estimation Algorithms

- CNN Regression Models
- Multi-task CNN
- Depth CNN
- Stereo Vision Models
- Structure from Motion (SfM)

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6. Nutrition Regression Algorithms

- Linear Regression
- Random Forest Regression
- CNN Regression
- Multi-output Neural Networks
- Dense Regression Networks

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7. Transformer-Based Algorithms

- Vision Transformer (ViT)
- Swin Transformer
- DeiT (Data-efficient Image Transformer)
- BEiT (Bidirectional Encoder Representation from Image Transformers)
- ConvNeXt

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8. Hybrid CNN-Transformer Models

- ResNet + ViT
- EfficientNet + ViT
- CNN + Swin Transformer
- ConvNeXt + Transformer

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9. Multimodal AI Models

- CLIP
- BLIP
- BLIP-2
- Flamingo
- LLaVA
- GPT-4V
- Gemini Vision
- Qwen-VL

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10. Large Language Models (LLMs)

- GPT-4
- GPT-4V
- Gemini
- Claude
- Llama 3
- DeepSeek
- Qwen

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11. Visual-Ingredient Fusion Models

- Visual-Ingredient Feature Fusion (VIF^2)
- Ingredient-aware Transformers
- Cross-modal Attention Networks

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12. Segment Anything Models

- SAM (Segment Anything Model)
- MobileSAM
- FastSAM
- MedSAM

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13. Self-Supervised Learning Algorithms

- SimCLR
- MoCo
- DINO
- BYOL
- MAE (Masked Autoencoders)

- iBOT

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14. Graph Neural Network Algorithms

- Graph Convolutional Network (GCN)
- GraphSAGE
- Graph Attention Network (GAT)

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15. Reinforcement Learning Algorithms

- Q-Learning
- Deep Q Networks (DQN)
- Proximal Policy Optimization (PPO)

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16. Sensor Fusion Algorithms

- Multimodal Neural Networks
- Attention Fusion Networks
- Ensemble Learning